

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

WEB-BASED MR. SOFTY INTEGRATED BUSINESS INTELLIGENCE PLATFORM

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In today's world, technology plays a very important role in how businesses operate. Many businesses are now using computerized systems to make their work faster, more accurate, and more organized. Instead of relying on manual processes, digital systems help in recording transactions, tracking inventory, and managing employee activities more efficiently.

For small and medium-sized businesses, using technology can make a big difference. It helps reduce human errors, saves time, and makes it easier to access important data. With the help of computerized systems, business owners can monitor their operations in real time and make better decisions based on accurate information.

However, not all businesses have fully adapted to this change. Some small businesses still rely on manual methods such as writing sales in notebooks, tracking attendance using logbooks, and monitoring inventory without a proper system. While these methods may still work for very small operations, they often become difficult to manage as the business grows.

Manual processes can lead to several problems. Records can be misplaced or damaged, calculations can be incorrect, and it can take a long time to retrieve information. These issues can affect the overall performance of the business. For example, inaccurate sales records may lead to wrong financial reports, while poor inventory tracking can cause stock shortages or excess inventory. In addition, manual attendance tracking may result in errors in employee work hours.

Mr. Softy, a local business, currently uses manual processes in handling sales recording, employee attendance, and inventory management. Because of this, it becomes difficult to keep track of daily transactions, monitor employee attendance properly, and maintain accurate inventory records. As the business continues to operate, these challenges may slow down operations and increase the chances of errors.

To solve these problems, this study proposes the development of a **Web-Based Mr. Softy Integrated Business Intelligence Platform**. The system is designed to automate important business processes such as sales recording, attendance tracking, and inventory management. By using this system, the business can improve accuracy, reduce manual work, and make

operations more efficient. It will also help the business owner access important data easily and make better decisions.

1.2 Statement of the Problem

Mr. Softy currently manages its sales, employee attendance, and inventory using manual methods. While this approach is simple, it creates several challenges that affect the efficiency and accuracy of the business operations. Manual processes are time-consuming, prone to errors, and make it difficult to access and organize data.

Another issue is that the business does not have a single system that connects all its operations. Because of this, information is scattered, and it becomes harder for the owner to monitor the business and make decisions based on reliable data.

With these concerns, this study aims to answer the following questions:

1. How can sales transactions be automated to reduce errors and improve the recording of daily sales?
2. How can employee attendance be monitored and recorded more accurately?
3. How can inventory be managed efficiently to avoid stock shortages and incorrect records?
4. How can integrating sales, attendance, and inventory into one system improve business operations?
5. How can the system help the business owner make better decisions?

1.3 Objectives of the Study

1.3.1 General Objectives

The main goal of this study is to develop a web-based system for Mr. Softy that will automate sales recording, employee attendance tracking, and inventory management to improve the overall efficiency and accuracy of the business.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. To develop a system that can automatically record daily sales transactions to make the process faster and more accurate.
2. To create a feature that can track employee attendance in real time and reduce errors in recording work hours.

3. To design an inventory module that can monitor product stocks and help prevent shortages or overstocking.
4. To reduce errors and delays caused by manual processes in handling business data.
5. To provide organized and accessible records that can help the business owner make better decisions.

1.4 Scope and Delimitation

This study focuses on developing a web-based system for Mr. Softy that will manage sales, employee attendance, and inventory. The system will allow authorized users, such as staff and administrators, to record transactions, track attendance, and monitor product stocks.

The system will include features such as order recording, attendance logging, inventory tracking, and basic reporting. It will be accessible through a web browser, making it easy to use within the business.

However, the study has some limitations. It will only focus on internal business operations and will not include features such as online payments, delivery tracking, or mobile applications. Advanced financial reports and large-scale system deployment are also not included in this study.

1.5 Significance of the Study

The results of this study are expected to benefit the following:

- **Mr. Softy Business** - The system will help the business manage sales records, employee attendance, and product inventory more efficiently.
- **Business Owner / Management** - The system will provide organized and accurate data that can help the owner monitor sales, employee attendance, and product stocks.
- **Employees** - The system will make attendance tracking easier and more accurate. It will help ensure that employees' work hours are properly recorded.
- **Customers** - Although the system is mainly for internal use, improved business operations can lead to better service and faster transactions for customers.
- **Future Researcher** - This study may serve as a reference for future researchers who plan to develop similar systems related to sales management, employee attendance monitoring, and inventory management.

1.6 Definition of Terms

Business Intelligence Platform - A system that analyzes business data to support smart business decisions.

Computerized Management System - A digital system used to manage and organize business operations.

Employee Attendance Tracking - The process of recording and monitoring employee work hours.

Inventory Management - The process of tracking and controlling product stock.

Manual System - A method of managing records without using automated computer systems.

Product Inventory - The products or goods available for sale in a business.

Sales Recording - The process of documenting sales transactions.

Stock Monitoring - The process of checking and updating product stock levels.

Web-Based System - A system that is accessed through a web browser using the internet or a network.

Authorized User - A person who has permission to access and use the system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Business Intelligence (BI) systems have become an essential tool for modern organizations seeking to improve decision-making and operational efficiency. According to the study conducted by Joaquim Lapa, Jorge Bernardino, and Ana Figueiredo (2014), BI platforms are designed to collect, process, and analyze large volumes of business data to generate meaningful insights. Their research compared several open-source BI tools such as Actuate, JasperSoft, OpenI, Palo, Pentaho, SpagoBI, and Vanilla. The study emphasized that selecting the appropriate BI platform depends on the specific needs of the organization, including scalability, usability, and data integration capabilities. The researchers concluded that BI systems significantly improve data accessibility and reporting accuracy, which are critical for business growth.

Another study by Wixom and Watson (2010) highlighted the role of Business Intelligence in enhancing organizational performance. Their findings revealed that companies that implement BI systems are more capable of making data-driven decisions, improving operational efficiency, and gaining competitive advantage. The study also pointed out that BI systems help reduce reliance on manual processes, thereby minimizing errors and improving productivity.

In addition, a study conducted by Chen, Chiang, and Storey (2012) discussed the evolution of business intelligence and analytics. The researchers explained that BI systems have transitioned from simple reporting tools to advanced analytics platforms capable of real-time data processing. Their findings suggest that businesses that adopt BI technologies are better equipped to respond to market changes and customer demands.

Moreover, Turban et al. (2011) examined the integration of BI systems in small and medium enterprises (SMEs). The study found that while large organizations have widely adopted BI technologies, SMEs often face challenges such as limited resources and lack of technical expertise. However, the study emphasized that implementing even basic BI solutions can significantly improve business operations and decision-making in SMEs.

A study by Laudon and Laudon (2016) focused on the importance of information systems in business operations. The researchers explained that computerized systems, including BI platforms, help organizations streamline processes such as sales tracking, inventory management, and employee monitoring. Their findings support the idea that transitioning from manual to automated systems enhances efficiency, accuracy, and data reliability.

Furthermore, a study conducted by O'Brien and Marakas (2011) highlighted the advantages of web-based information systems. The researchers stated that web-based systems allow users to access information anytime and anywhere, making them highly suitable for businesses that require real-time data access. The study concluded that web-based platforms improve collaboration, data sharing, and overall system flexibility.

Another relevant study by Stair and Reynolds (2018) discussed the impact of information systems on business efficiency. Their research emphasized that automated systems reduce human errors, speed up transaction processing, and improve data management. The study also noted that integrating multiple business functions into a single system leads to better coordination and decision-making.

2.2 Local Studies

In the Philippine context, several studies have explored the adoption of computerized systems among small and medium enterprises (SMEs). A study titled "Inventory and Point-of-Sale System for Bida Mart" revealed that many local businesses still rely on manual processes for managing sales and inventory. The study found that manual systems often lead to inaccurate records, stock shortages, and inefficient operations. The researchers recommended the implementation of automated systems to improve accuracy and efficiency.

Another study by Barroga, Wabina, and Sales (2021) examined the level of technology adoption among micro, small, and medium enterprises (MSMEs) in the Philippines. Their findings indicated that many MSMEs have limited awareness of advanced technologies such as business intelligence systems. The study also highlighted challenges such as financial constraints, lack of technical skills, and resistance to change, which hinder the adoption of digital solutions.

A study conducted by Dela Cruz and Santos (2020) focused on the development of a web-based inventory system for small businesses. The researchers found that implementing a web-based system significantly improved inventory tracking, reduced errors, and enhanced overall business efficiency. The study emphasized that web-based solutions are cost-effective and accessible, making them suitable for small businesses.

Furthermore, a study by Reyes et al. (2019) explored the use of attendance monitoring systems in local businesses. The study revealed that manual attendance tracking often results in inaccurate records and difficulty in monitoring employee performance. The researchers concluded that automated attendance systems improve accuracy, reduce administrative workload, and enhance employee accountability.

Another local study by Garcia and Lopez (2022) examined the implementation of integrated business systems in retail businesses. The study found that integrating sales, inventory, and reporting systems into a single platform improves data consistency and operational efficiency. The researchers emphasized the importance of system integration in supporting business growth and decision-making.

2.3 Related Literature

The adoption of digital technologies in business operations has become increasingly important, particularly for micro, small, and medium enterprises (MSMEs). According to the Philippine Institute for Development Studies (PIDS, 2025), many SMEs are aware of the benefits of digital transformation but face challenges such as high implementation costs, limited technical knowledge, and resistance to change. These barriers prevent businesses from fully utilizing digital tools to improve efficiency and competitiveness.

Inventory management systems are essential for businesses that handle physical products.

According to Stevenson (2018), inventory systems help businesses track stock levels, manage supply chains, and prevent stock shortages. Automated inventory systems provide real-time updates, reducing the risk of overstocking or understocking.

Point-of-sale (POS) systems play a critical role in sales management. Laudon and Laudon (2016) explained that POS systems automate the recording of sales transactions, generate receipts, and provide sales reports. These systems improve transaction speed, accuracy, and customer service.

Attendance monitoring systems are also important for managing employees. According to Mathis and Jackson (2011), automated attendance systems improve the accuracy of employee records and reduce the time required for administrative tasks. These systems also help ensure compliance with company policies and labor regulations.

Web-based systems have become widely used due to their accessibility and flexibility.

According to Shelly and Rosenblatt (2012), web-based applications allow users to access systems through a web browser without the need for installation. This makes them cost-effective and easy to maintain.

Business Intelligence systems integrate data from different sources to provide meaningful insights. Turban et al. (2011) stated that BI systems help organizations analyze data, identify trends, and make informed decisions. These systems are particularly useful for businesses that need to monitor performance and improve operations.

2.4 Synthesis of Reviewed Literature

The reviewed literature and studies highlight the growing importance of digital technologies in improving business operations. Foreign studies emphasize the role of business intelligence systems, web-based platforms, and integrated information systems in enhancing efficiency, accuracy, and decision-making. These studies demonstrate that automated systems reduce manual work, minimize errors, and provide real-time access to data.

Local studies, on the other hand, reveal that many MSMEs in the Philippines still rely on manual processes, which lead to inefficiencies and inaccurate records. The studies also highlight challenges such as limited resources, lack of technical expertise, and resistance to change. Despite these challenges, the findings suggest that implementing web-based and integrated systems can significantly improve business performance.

The literature also shows that there is a gap in the availability of affordable and user-friendly digital solutions tailored for small businesses. While many systems exist, they are often not accessible or suitable for SMEs. This gap highlights the need for a system that integrates sales, inventory, and attendance management into a single platform.

Therefore, this study aims to develop a Web-Based Mr. Softy Integrated Business Intelligence Platform that addresses these challenges by providing an efficient, accurate, and user-friendly system for managing business operations. The proposed system is expected to improve data management, reduce manual workload, and support better decision-making for the business.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study uses a **developmental research design** to develop a web-based integrated business intelligence platform for Mr. Softy. This design focuses on the creation, design, and evaluation of a system intended to improve business operations, including inventory management, sales tracking, and data analysis. The system will be developed based on identified problems and user requirements to ensure efficiency and usability.

3.2 System Development Methodology

This study will use the **Agile methodology** in developing the web-based integrated business intelligence platform for Mr. Softy, consisting the following phases:

1. Requirements Analysis
2. System Design
3. Development
4. Testing
5. Deployment (Prototype)
6. Continuous Improvement

3.3 System Architecture

The **Mr. Softy Web-Based Integrated Business Intelligence Platform** follows a **client-server architecture**. In this architecture, the **users (clients)** interact with the system through a **web or mobile interface**, while the **server** handles all **data processing, business logic, machine learning predictions, and database management**. This structure ensures efficient processing, centralized data storage, and secure access for multiple users simultaneously.

3.4 Tool and Technology Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP, MySQL
- **Database:** MySQL Work Bench
- **Development Tools:** Visual Studio Code

3.5 Data Gathering Techniques

- **Interviews**
- **Observation**

3.6 Testing Procedures

- **Unit Testing-** Test individual parts of the system.
- **Integration Testing-** Test how different parts work together.
- **System Testing-** Test the whole system as one.
- **User Acceptance Testing-** Let the user/client test the system.

3.7 Evaluation Criteria

- **Functionality-** Features work correctly and perform intended tasks.
- **Usability-** The system is easy to use and navigate.
- **Reliability-** The system operates consistently without crashing.
- **Security-** Data is protected and inputs are validated.
- **Efficiency-** The system responds quickly and uses resources effectively.

SUPPORTING DIAGRAMS

DATA FLOW DIAGRAM

User → Mr. Softy System → Database

- **Staff** interacts with the system to input customer orders and payments.
- **The system** handles order processing, inventory updates, and receipt generation.
- **Database** stores all order, payment, and product information.
- **Admin** manages inventory and accesses reports; analytics are generated and stored for decision-making.

ENTITY RELATIONSHIP DIAGRAM

Entities:

- **User**(UserID, Name, Username, Password, StoreID, Role)
- **Store**(StoreID, StoreName, Location)
- **Inventory**(InventoryID, StoreID, ItemID, Quantity, LowStockLevel, UpdatedAt)
- **Products**(ProductID, ProductName, Size, Price, Category, Status, CreatedAt)
- **ToppingInventory**(ToppingInventoryID, BranchID, ToppingID, Quantity, LowStockLevel, UpdatedAt)
- **StockRequest**(RequestID, StoreID, ItemID, RequestedQty, Status, RequestedBy, CreateAt)
- **Item**(ItemID, ItemName, Category)
- **ProductItems**(ProductItemID, ProductId, ItemID, QuantityNeeded)
- **Notification**(NotifID, UsedID, StoreID, RequestID, ItemName, RequestedQty, Status, IsRead, CreatedAt, UpdatedAt)
- **StockCheckLogs**(LogID, StoreID, StaffID, ItemID, CheckedAt, Remarks, CheckDate)
- **Attendance**(AttendancelD, UserID, StoreID, Role, Date, TimeIn, TimeOut)
- **Toppings**(ToppingID, ToppingName, Price, Status, CreatedAt)
- **Sales**(SaleID, ProductID, CashierID, StoreID, Quantity, Subtotal, SaleDate)
- **SaleToppings**(SaleToppingID, SaleID, ToppingID)
- **Orders**(OrderID, CashierID, StoreID, TotalAmount, OrderDate, OrderTime, CreatedAt)

Relationships:

- A **User** works at a **Store**.
- A **Store** has **Items** in its **Inventory** and **Toppings** in **ToppingInventory**.
- A **Product** is made from **Items** (through **ProductItems**).
- A **User** can request more **Items** via a **StockRequest**, which can trigger a **Notification**.
- **Users** can log **StockChecks** and **Attendance** at a **Store**.
- A **Sale** happens when a **User** sells a **Product** at a **Store**.
- **SaleToppings** link **Sales** to the **Toppings** sold with a **Product**.
- An **Order** contains multiple **Sales** and happens at a **Store**.

USE CASE DIAGRAM

Actors:

- **Staff / User**
- **Admin**

Use Cases:

Staff / User

- Log In to the System
- Add Orders for Customers (Products + Toppings)
- View and Manage Orders
- Void Incorrect Items
- Process Checkout and Billing
- Check Inventory
- Create Stock Requests
- Log Stock Checks
- Record Attendance

Administrator:

- Log In to the System
- Manage Products and Toppings
- View Sales Reports
- Monitor Transactions and Orders
- Manage User Accounts
- Oversee Store Operations and Notifications